SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.9: Linear Motion — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.9: Linear Motion
Driving Question	DRIVING QUESTION: How can we use mathematics — calculations and graphs — to fully describe and explain the motion of any moving body, from a Nairobi Marathon runner to a matatu on the Nakuru highway?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon & Key Terms: What Is Really Happening in the Nairobi Marathon?**	Learners examined a simplified race report of a Nairobi Marathon runner — including split times, route map segments, and a finish-line photo — and discussed what they noticed and wondered about the runner's journey. They compared the total road distance run (42.195 km of winding Nairobi streets) with the straight-line displacement from Nyayo Stadium start to finish, observing that the two values are not equal.	Distance is the total path length covered, measured in metres or kilometres, and is a scalar quantity — direction is not required. Displacement is the straight-line change in position from start to end point, measured in the same units but also requiring a stated direction, making it a vector quantity. Speed is the rate of change of distance (distance ÷ time); velocity is the rate of change of displacement (displacement ÷ time) and also requires direction. These four terms are the foundational vocabulary for all subsequent lessons.	The Nairobi Marathon runner covers more distance than their displacement because the road curves — total tarmac underfoot ≠ straight-line gap between start and finish. Pedagogical note: stress early that confusing 'distance' and 'displacement' is the single most common error in this sub-strand. Use a physical demonstration — walk a curved path around desks, then stretch a string straight back — to make the distinction concrete before any calculation is attempted. Cold-call at least four learners to state a definition in their own words before moving on.
Lesson 2 **Calculating Speed, Velocity & Time: Putting Numbers to the Race**	Learners were given a structured data table containing split distances and split times for three fictional runners in the Nairobi Marathon (Runner A — Eliud from Eldoret, Runner B — Amina from Mombasa, Runner C — Otieno from Kisumu). They performed calculations for each segment: average speed = distance ÷ time, average velocity = displacement ÷ time, and rearranged the formula to find missing distance or time values. A matatu scenario on the Nakuru	The mathematical relationships speed = distance/time, velocity = displacement/time, distance = speed × time, and time = distance/speed can be rearranged and applied to solve for any one unknown given the other two. Average speed and instantaneous speed are distinguished: average speed covers the whole journey; instantaneous speed is the speed at one particular moment. Units must be consistent — convert km/h to m/s by dividing by 3.6 (or	Runner A may have a higher average speed over the first 10 km than Runner B, yet Runner B finishes faster overall — demonstrating that a single-segment calculation cannot describe an entire journey. Pedagogical note: insist learners write the formula, substitute values with units, and write a concluding statement with units in every worked example. A common error is dividing time by distance instead of the reverse — build a 'formula triangle'

	highway was introduced as a parallel context.	multiplying by 5/18).	anchor chart for the classroom wall. Provide at least two km/h ↔ m/s conversion practice items before the matatu problem.
Lesson 3 **Drawing Displacement-Time Graphs: Reading the Shape of the Race**	Learners plotted displacement-time (s-t) graphs for Eliud's marathon segments using pre-prepared data tables, choosing appropriate scales and labelling axes correctly. They then compared their graphs with a teacher-drawn reference graph projected on the board, identifying where the line was steep, gentle, horizontal, or sloping downward.	The shape of a displacement-time graph tells the full story of motion: a steep positive line indicates high speed in the positive direction; a gentle positive line indicates low speed; a horizontal line (zero gradient) means the object is stationary; a negative slope means the object is moving back toward the origin. The graph is a visual narrative — each segment's shape corresponds directly to a phase of the journey.	Eliud's s-t graph rises steeply in the early kilometres (fresh legs, downhill from Uhuru Highway), flattens slightly on the return loop (fatigue, uphill), and continues to rise to the finish — the varying steepness encodes the changing pace. Pedagogical note: before learners plot, spend 3 minutes on axis-scaling decisions — a poorly scaled graph is the most frequent source of error. Ask learners to describe each segment in plain English BEFORE they attempt to calculate gradients, so graph-reading precedes gradient arithmetic. This lesson is the bridge between numerical and graphical representations.
Lesson 4 **Interpreting Displacement-Time Graphs: Gradient as Velocity**	Learners calculated the gradient of each straight-line segment on their previously drawn s-t graphs using gradient = rise ÷ run = $\Delta s \div \Delta t$, then compared the calculated gradient values to the velocity values computed numerically in Lesson 2. They also examined a provided graph of the Nakuru matatu journey and identified stationary periods, high-speed segments, and direction reversals.	The gradient of a displacement-time graph is numerically equal to the velocity of the object during that segment. A zero gradient equals zero velocity (object is stationary). A negative gradient equals motion in the negative direction (back toward or past the start). A steeper gradient equals greater speed. This is the key quantitative link between the graphical and numerical representations.	Learners can now extract a velocity value directly from a graph without a separate data table — the graph itself is a calculation tool. Pedagogical note: emphasise that gradient is always calculated as vertical change ÷ horizontal change ($\Delta y \div \Delta x$), never the reverse. A persistent misconception is that a 'steeper' line always means 'farther from start' — clarify that steepness encodes rate, not position. Use at least two cold-call 'what is the velocity in this segment?' questions where the answer is zero or negative to surface and correct this error.
Lesson 5 **Introduction to Velocity-Time Graphs: Drawing the Matatu's Speed Story**	Learners were given a new data set describing a matatu journey from Nairobi's CBD to Nakuru: several phases including acceleration from rest at a bus stage, constant cruising speed on the open highway, deceleration into Naivasha town, a stationary stop, and further acceleration to Nakuru. Learners plotted velocity-time (v-t) graphs from this data, choosing appropriate scales and labelling axes with correct units.	A velocity-time graph plots velocity (y-axis, m/s or km/h) against time (x-axis, seconds or minutes). A horizontal line means constant velocity (no acceleration). A positive slope means the object is accelerating (speeding up). A negative slope means the object is decelerating (slowing down). A line touching the x-axis means the object is at rest ($v = 0$). The v-t graph is a complementary tool to the s-t graph — together they give a complete picture of motion.	The matatu's v-t graph rises from zero (starting from rest at the Nairobi bus stage), levels off at a constant cruise velocity on the open highway, dips to zero at the Naivasha stop, then rises again — the graph shape matches the real physical experience of the journey. Pedagogical note: many learners initially confuse the s-t and v-t graph shapes. Display both graph types side by side on the board throughout this lesson. A useful teaching move is to ask learners to physically stand up and mime acceleration (lean forward), constant

			speed (walk steadily), and deceleration (lean back) while describing each graph segment.
Lesson 6 **Interpreting Velocity-Time Graphs: Area Under the Graph as Distance**	Learners examined a velocity-time graph for the matatu journey and were guided to calculate: (i) the gradient of each sloped segment to find acceleration or deceleration values, and (ii) the area of each geometric shape under the graph (rectangles for constant speed, triangles for acceleration/deceleration phases) to find distance travelled in each phase.	The gradient of a velocity-time graph equals acceleration: $a = \Delta v \div \Delta t$. A steep positive gradient means rapid acceleration; a negative gradient (deceleration) has a negative value. The area under a velocity-time graph equals the distance travelled during that time interval. For rectangular sections: area = base × height = time × velocity = distance. For triangular sections (acceleration/deceleration): area = ½ × base × height = ½ × time × Δv = distance. The unit check confirms this: (m/s) × s = m.	The matatu's total journey distance can be recovered entirely from the v-t graph by summing all the areas under each segment — no separate distance data is needed. Pedagogical note: the area-under-the-graph concept is the most conceptually demanding idea in this sub-strand. Use colour-coded shading for each geometric region before any calculation. Insist learners label each shape (rectangle, triangle), write the area formula, substitute, and state units. A frequent error is using the full height of a trapezoid as the triangle height — decompose trapezoids into rectangle + triangle explicitly.
Lesson 7 **Equations of Linear Motion & Combining Graphs: Solving the Full Journey**	Learners were introduced to the four equations of linear motion ($v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$; $s = \frac{1}{2}(u + v)t$) and applied them to solve multi-step problems set in Kenyan contexts — a boda-boda accelerating from rest in Kisumu, a Rift Valley Express bus braking on the Nakuru highway, and an athlete's sprint at the Moi International Sports Centre. Learners also practised moving between graph and equation representations for the same scenario.	The four kinematic equations apply to objects undergoing uniform (constant) acceleration in a straight line. Each equation links a different combination of the five variables: initial velocity (u), final velocity (v), acceleration (a), time (t), and displacement (s). The strategy for solving any problem is: (1) list known and unknown quantities with units, (2) select the equation that contains the unknown and all three known quantities, (3) substitute and solve, (4) state the answer with correct units and direction where applicable.	The equations and the graphs are two languages describing the same physical reality — for example, $v = u + at$ is the algebraic form of reading the gradient on a v-t graph, and $s = \frac{1}{2}(u + v)t$ is the algebraic form of calculating the trapezoid area. Pedagogical note: display the four equations permanently on the board for the remainder of the unit. A structured 'KUSD' framework (Known, Unknown, Select equation, Do the maths) reduces errors significantly. Require learners to check answers by substituting back into the original equation before writing a final answer.
Lesson 8 **Consolidation, Model Building & Final Explanation: The Full Story of the Nairobi Marathon**	Learners returned to the Nairobi Marathon anchoring phenomenon and used all tools acquired across Lessons 1–7 to produce a complete mathematical explanation of the race. They calculated average speed and velocity for each race segment, identified acceleration and deceleration phases, drew and annotated both s-t and v-t graphs for the full 42.195 km journey, computed distances from graph areas, and applied kinematic equations to answer 'what if' extension questions about the race.	A moving body's complete motion can be described using: scalar quantities (distance, speed) and vector quantities (displacement, velocity, acceleration); displacement-time graphs (gradient = velocity); velocity-time graphs (gradient = acceleration, area = distance); and the four kinematic equations for uniform acceleration. These tools are complementary — numerical, graphical, and algebraic representations all encode the same underlying motion and can be used interchangeably depending on what	The Nairobi Marathon runner's journey is now fully described mathematically: the winding road gives total distance 42.195 km; the net displacement from Nyayo Stadium start to finish line is calculable from the route map; each split segment has a calculable average velocity; acceleration and deceleration phases are visible on the v-t graph and quantifiable via kinematic equations. Pedagogical note: this lesson functions as a summative performance task. Learners should work in pairs to

		information is given or required.	produce a poster or annotated graph set — assess both the mathematical accuracy and the quality of the written explanation linking calculations back to the driving question. Celebrate the journey: learners began by observing a race; they now have the full mathematical toolkit to describe any moving body.